SynthMicro — Whole Blood Smear XAI Report

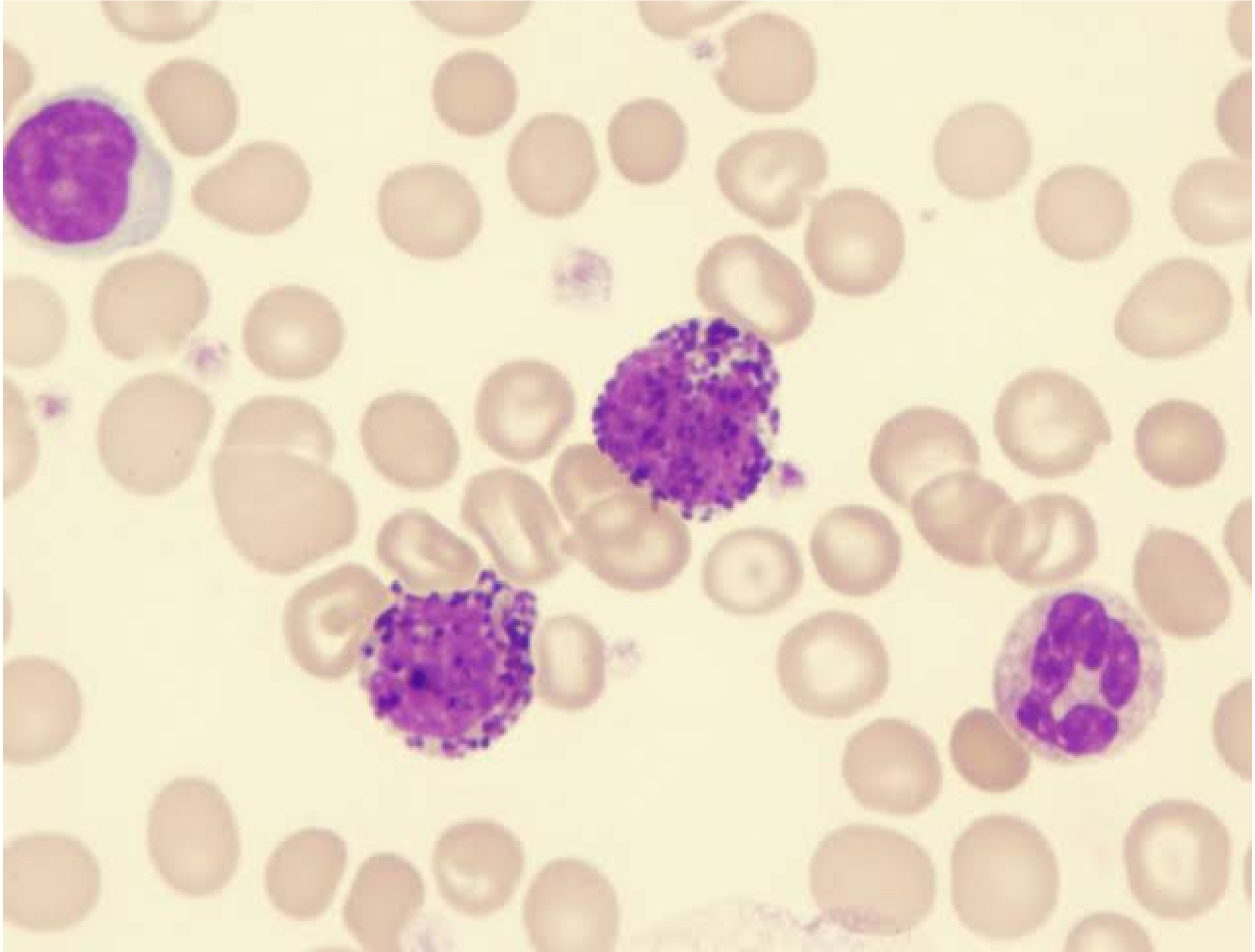
Analysis ID: 053a89b5f7a840b1bbbf29d1d2033152
Generated: 2026-09-07 22:53:23

Pipeline: Cellpose segmentation → 384×384 cell extraction → ResNet50 classification → Grad-CAM → SHAP → explanatory report.

Measurement	Result
Cellpose objects	57
Nucleated-cell candidates	4
Myeloblast-like candidates	3
Myeloblast-like proportion	75.00%

Screening interpretation: the proportion above is a research-oriented model indicator based on candidate cells selected by the notebook's purple-score rule. It is not a cancer probability and does not establish AML or another diagnosis.

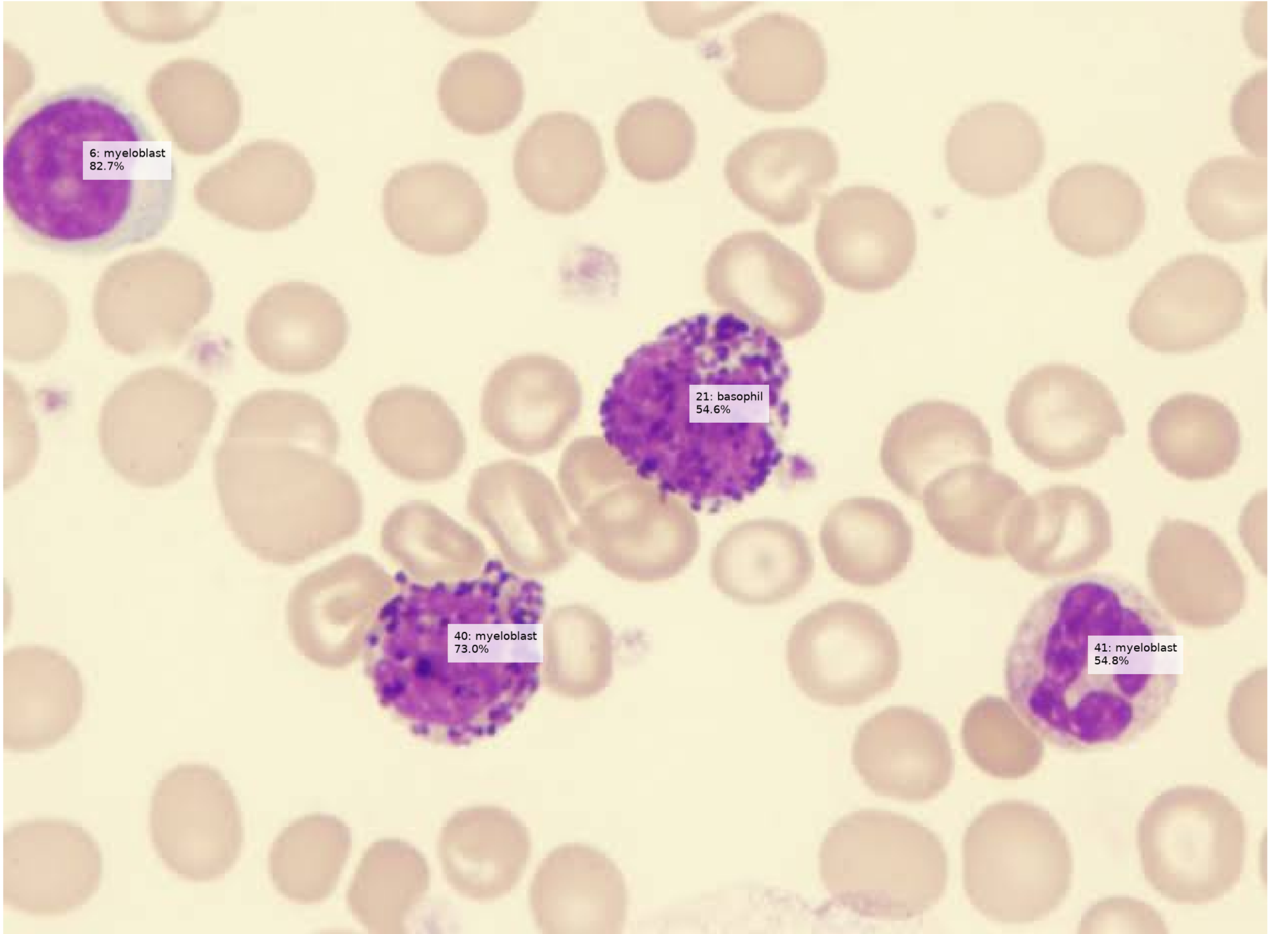
Whole Smear



Original uploaded blood-smear image analyzed by Cellpose.

Segmentation and Candidate Overlay

Whole Smear — Candidate Cell Classification



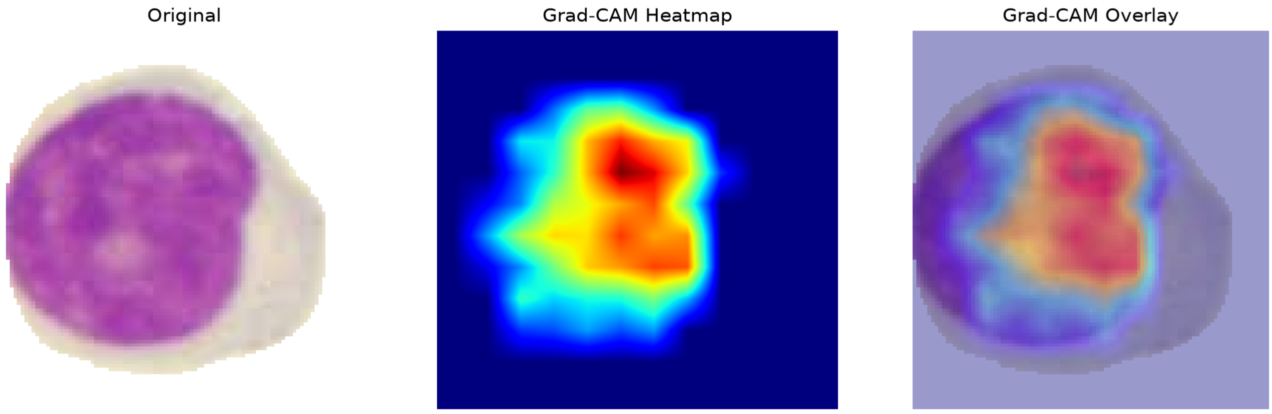
Cell-Level Results

ID	Area	Purple	Class	Confidence
6	6145	58.8	myeloblast	82.7%
21	7338	76.9	basophil	54.6%
40	6570	72.8	myeloblast	73.0%
41	6330	54.1	myeloblast	54.8%

Grad-CAM Explanations

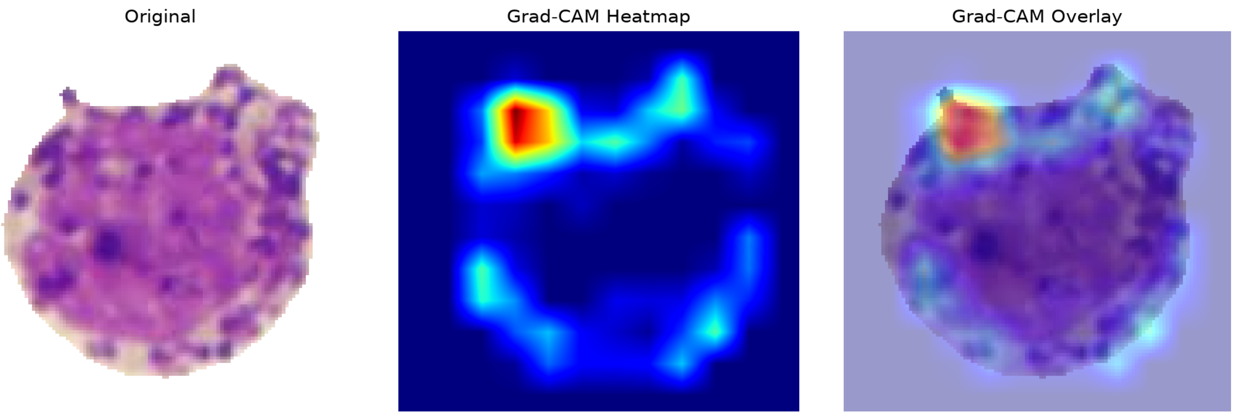
Grad-CAM shows regions contributing to the selected model class. It is an explanation of model behavior, not a direct measurement of a biological structure.

Cell 6 — myeloblast (82.7%)



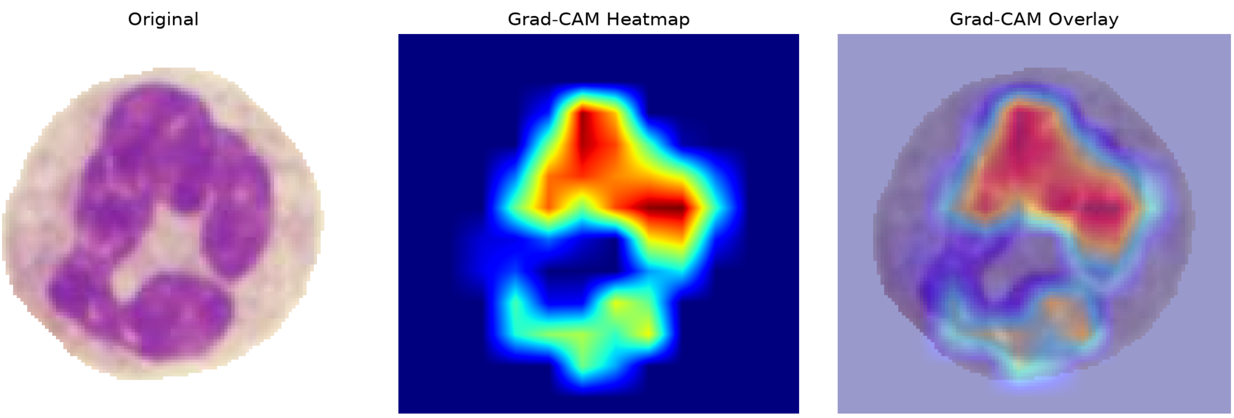
The ResNet50 model classified this segmented candidate as a myeloblast with 82.7% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 40 — myeloblast (73.0%)



The ResNet50 model classified this segmented candidate as a myeloblast with 73.0% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 41 — myeloblast (54.8%)



The ResNet50 model classified this segmented candidate as a myeloblast with 54.8% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

SHAP Explanations

SHAP attribution magnitude summarizes the magnitude of pixel-level contribution toward the selected class. It is a model explanation, not a clinical measurement.

SHAP output was unavailable for this analysis.

Interpretation and Limitations

- The classifier has five closed-set classes: basophil, erythroblast, monocyte, myeloblast and segmented neutrophil.
- RBCs and unknown cell types are not explicit classes and can therefore be forced into one of the five outputs.
- Cellpose segmentation quality affects every downstream result.
- Stain, microscope, illumination and acquisition differences can cause domain shift.
- Confidence is not a calibrated probability of disease.
- A myeloblast-like prediction is not equivalent to an AML diagnosis.
- Clinical assessment requires appropriate hematopathology and laboratory testing.